I – Probability Density Functions

a) Variables

Definition

For a continuous random variable X, x can take any real value. We use a function called a probability density function to define the probability distribution. Probabilities are found by calculating areas under the probability density function curve for a particular interval.

Definition

For a continuous random variable X, the *probability density function* is a function $f(x)$ such that $f(x) \geqslant 0$ on its entire domain.

If the domain of the function is $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx = 1$.

The probability that X lies in the interval $c \leqslant X \leqslant d$ is

$$\mathrm{P}(c \leqslant X \leqslant d) = \int_c^d f(x)\,dx.$$

Note. *For a continuous variable X, the probability that X is exactly equal to a particular value is zero. So $\mathrm{P}(X = x) = 0$ for all x.*

So, for a continuous variable we can only talk about the probability that an event lies in an interval. A consequence of this is that

$$\mathrm{P}(c \leqslant X \leqslant d) = \mathrm{P}(c < X \leqslant d) = \mathrm{P}(c \leqslant X < d) = \mathrm{P}(c < X < d).$$

This would not be true if X was discrete.

II – The Normal Distribution

a) Historical note

The normal distribution was first characterised by Carl Friedrich Gauss in 1809 as a way to rationalize his method of least squares for linear regression.

Marquis de Laplace was the first to calculate $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$. This led to the correct normalization constant $\frac{1}{\sigma\sqrt{2\pi}}$ being used for the normal distribution, ensuring $\int_{-\infty}^{\infty} f(x)\,dx = 1$.

b) Some examples

The normal distribution is the most important distribution for a continuous random variable. Many naturally occurring phenomena have a distribution that is normal, or approximately normal. Some examples are :

— physical attributes of a population such as height, weight, and arm length

— crop yields

— scores for tests taken by a large population.

Once a normal model has been established, we can use it to make predictions about a distribution and to answer other relevant questions.

Consider the oranges picked from an orange tree. They do not all have the same weight. The variation may be due to several factors, including :

— genetics

— different times when the flowers were fertilised

— different amounts of sunlight reaching the leaves and fruit

— different weather conditions such as the prevailing winds.

The result is that most of the fruit will have weights close to the mean, while fewer oranges will be *much* heavier or *much* lighter. This results in a bell-shaped distribution for the weight of oranges in a crop, which is symmetric about the mean.

c) Definition

Definition

If X is normally distributed then its probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \qquad \text{for} \quad -\infty < x < \infty$$

where μ is the mean and σ^2 is the variance of the distribution.

We write $X \sim \mathrm{N}(\mu,\, \sigma^2)$.

Property

— The curve $y = f(x)$, called a normal curve, is symmetrical about the vertical line $x = \mu$.

— As $x \to \pm\infty$ the normal curve approaches its asymptote, the x-axis.

— $f(x) > 0$ for all x.

— Since the total probability must be 1, $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$.

— More scores are distributed closer to the mean than further away. This results in the typical bell shape.

d) Points of inflection

Differentiating $f(x) = \dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$, we obtain

$$f'(x) = \frac{-1}{\sigma^2\sqrt{2\pi}}\left(\frac{x-\mu}{\sigma}\right) e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\therefore\ f'(x) = 0$ only when $x = \mu$, and this corresponds to the maximum of $f(x)$.

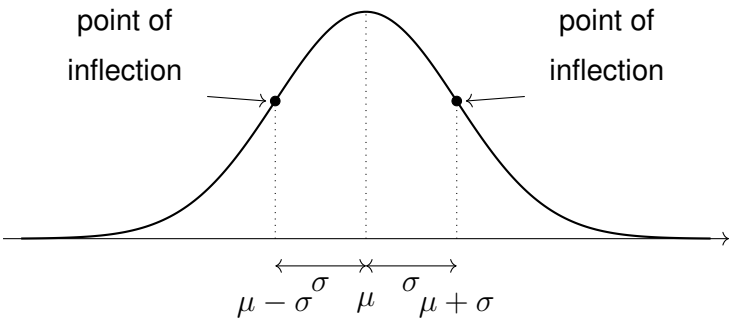
Differentiating again, we obtain

$$f''(x) = \frac{-1}{\sigma^2\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\left[\frac{1}{\sigma} - \frac{(x-\mu)^2}{\sigma^3}\right]$$

$\therefore\ f''(x) = 0$ when $\dfrac{(x-\mu)^2}{\sigma^3} = \dfrac{1}{\sigma}$

$$\therefore \quad (x-\mu)^2 = \sigma^2$$
$$\therefore \quad x - \mu = \pm\sigma$$
$$\therefore \quad x = \mu \pm \sigma$$

So, the points of inflection are at $x = \mu + \sigma$ and $x = \mu - \sigma$.

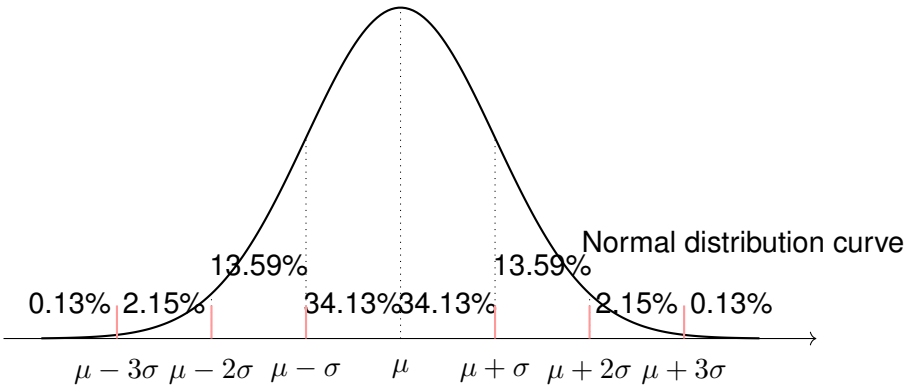


Definition

For a normal curve, the standard deviation is uniquely determined as the horizontal distance from the line of symmetry $x = \mu$ to a point of inflection.

III – Percentage and curve

For a normal distribution with mean μ and standard deviation σ, the percentage breakdown of where the random variable could lie is shown below.



Property

— $\approx 68.26\%$ of values lie between $\mu - \sigma$ and $\mu + \sigma$

— $\approx 95.44\%$ of values lie between $\mu - 2\sigma$ and $\mu + 2\sigma$

— $\approx 99.74\%$ of values lie between $\mu - 3\sigma$ and $\mu + 3\sigma$

Example

The chest measurements of 18 year old male footballers are normally distributed with a mean of 95 cm and a standard deviation of 8 cm.

1. Find the percentage of footballers with chest measurements between :
   (a) 87 cm and 103 cm
   (b) 103 cm and 111 cm
2. Find the probability that the chest measurement of a randomly chosen footballer is between 87 cm and 111 cm.
3. Find the value of k such that approximately 16% of chest measurements are below k cm.

EXERCISE 1

The mean height of players in a basketball competition is 184 cm. If the standard deviation is 5 cm, what percentage of them are likely to be :

1. taller than 189 cm
2. taller than 179 cm
3. between 174 cm and 199 cm
4. over 199 cm tall ?

EXERCISE 2

The mean average rainfall of Claudona for August is 48 mm with a standard deviation of 6 mm. Over a 20 year period, how many times would you expect there to be less than 42 mm of rainfall during August in Claudona ?

EXERCISE 3

Two hundred lifesavers competed in a swimming race. The mean time was 10 minutes 30 seconds, and the standard deviation was 15 seconds. Estimate the number of competitors who :

1. took longer than 11 minutes
2. took less than 10 minutes 15 seconds
3. completed the race in a time between 10 min 15 s and 10 min 45 s.

EXERCISE 4

The weights of Jason's oranges are normally distributed. 84% of the crop weigh more than 152 grams and 16% weigh more than 200 grams.

1. Find μ and σ for the crop.
2. What percentage of the oranges weigh between 152 grams and 224 grams ?

IV – The Standard Normal Distribution (Z-distribution)

a) Definition

Every normal X-distribution can be transformed into the standard normal distribution or Z-distribution using the transformation $z = \dfrac{x - \mu}{\sigma}$.

Investigation

Suppose a random variable X is normally distributed with mean μ and standard deviation σ. For each value of x we can calculate a z-score using the algebraic transformation $z = \dfrac{x - \mu}{\sigma}$.
What to do :

1. Consider the x-values : $1, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 4, 5, 5, 5, 5, 6, 6, 7$.
   (a) Draw a graph of the distribution to check that it is approximately normal.
   (b) Find the mean μ and standard deviation σ for the distribution of x-values.
   (c) Use the transformation $z = \dfrac{x - \mu}{\sigma}$ to convert each x-value into a z-value.
   (d) Find the mean and standard deviation for the distribution of z-values.

Definition

The Z-distribution has mean 0 and standard deviation 1. In fact, $Z \sim \mathrm{N}(0,\, 1)$.
The probability density function for the Z-distribution is

$$f(z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}z^2}, \qquad -\infty < z < \infty.$$

If x is an observation from a normal distribution with mean μ and standard deviation σ, the z-score of x,

$$z = \frac{x - \mu}{\sigma},$$

is the number of standard deviations x is from the mean.

For example :
— if $z = 1.84$, then x is 1.84 standard deviations to the right of the mean
— if $z = -0.273$, then x is 0.273 standard deviations to the left of the mean.

b) Using the Z-distribution

For $X \sim \mathrm{N}(\mu,\, \sigma^2)$ and $Z \sim \mathrm{N}(0,\, 1)$, and for any $x_1,\, x_2 \in \mathbb{R}$, $x_1 < x_2$, with corresponding z-values $z_1 = \dfrac{x_1 - \mu}{\sigma}$ and $z_2 = \dfrac{x_2 - \mu}{\sigma}$:

$$\mathrm{P}(X \geqslant x_1) = \mathrm{P}(Z \geqslant z_1)$$

z-scores are particularly useful when comparing two populations with different μ and σ. However, these comparisons will only be reasonable if both distributions are approximately normal.

Example

Kelly scored 73% in History where the class mean was 68% and the standard deviation was 10.2%. In Mathematics she scored 66%, where the class mean was 62% and the standard deviation was 6.8%. In which subject did Kelly perform better compared with the rest of her class ? Assume the scores for both subjects were normally distributed.

Example

If $X \sim \mathrm{N}(10,\, 2.3^2)$, find these probabilities :

1. $\mathrm{P}(8 \leqslant X \leqslant 11)$
2. $\mathrm{P}(X \leqslant 12)$
3. $\mathrm{P}(X > 9)$. Illustrate your results.

Example

In 1972 the heights of rugby players were approximately normally distributed with mean 179 cm and standard deviation 7 cm. Find the probability that a randomly selected player in 1972 was :

1. at least 175 cm tall
2. between 170 cm and 190 cm.

EXERCISE 5

Given $X \sim \mathrm{N}(\mu, \sigma^2)$ and $Z \sim \mathrm{N}(0, 1)$, determine the values of a and b such that :

1. $\mathrm{P}(\mu - \sigma < X < \mu + 2\sigma) = \mathrm{P}(a < Z < b)$
2. $\mathrm{P}(\mu - 0.5\sigma < X < \mu) = \mathrm{P}(a < Z < b)$
3. $\mathrm{P}(0 \leqslant Z \leqslant 3) = \mathrm{P}(\mu - a\sigma \leqslant X \leqslant \mu + b\sigma)$

Example

Use technology to illustrate and calculate :

1. $\mathrm{P}(-0.41 \leqslant Z \leqslant 0.67)$
2. $\mathrm{P}(Z \leqslant 1.5)$
3. $\mathrm{P}(Z > 0.84)$

EXERCISE 6

If Z is the standard normal distribution, find the following probabilities using technology. In each case sketch the regions.

1. $\mathrm{P}(-0.86 \leqslant Z \leqslant 0.32)$
2. $\mathrm{P}(-2.3 \leqslant Z \leqslant 1.5)$
3. $\mathrm{P}(Z \leqslant 1.2)$
4. $\mathrm{P}(Z \leqslant -0.53)$
5. $\mathrm{P}(Z \geqslant 1.3)$
6. $\mathrm{P}(Z \geqslant -1.4)$
7. $\mathrm{P}(Z > 4)$

EXERCISE 7

Max's customers put money for charity into a collection box on the front counter of his shop. The average weekly collection is approximately normally distributed with mean \$40 and standard deviation \$6. What proportion of weeks would he expect to collect :

1. between \$30.00 and \$50.00
2. at least \$50.00 ?

EXERCISE 8

Eels are washed onto a beach after a storm. Their lengths have a normal distribution with mean 41 cm and variance 11 cm^2.

1. If an eel is randomly selected, find the probability that it is at least 50 cm long.
2. Find the proportion of eels measuring between 40 cm and 50 cm long.
3. How many eels from a sample of 200 would you expect to measure at least 45 cm in length ?

EXERCISE 9

The speed of cars passing the supermarket is normally distributed with mean $56.3\ \mathrm{km\,h^{-1}}$ and standard deviation $7.4\ \mathrm{km\,h^{-1}}$. Find the probability that a randomly selected car has speed :

1. between 60 and 75 $\mathrm{km\,h^{-1}}$
2. at most 70 $\mathrm{km\,h^{-1}}$
3. at least 60 $\mathrm{km\,h^{-1}}$.

V – Inverse Normal

To answer questions of the form "find k such that $\mathrm{P}(X \leqslant k) = p$", the number k is known as a quantile. When finding quantiles we are given a probability and are asked to calculate the corresponding measurement. This is the *inverse* of finding probabilities, and we use the inverse normal function on our calculator.

Example

If $X \sim \mathrm{N}(23.6,\ 3.1^2)$, find k for which $\mathrm{P}(X < k) = 0.95$.

X has mean 23.6 and standard deviation 3.1. If $\mathrm{P}(X < k) = 0.95$ then $k \approx 28.7$.

Example

If Z is the standard normal distribution, find k such that $\mathrm{P}(Z > k) = 0.73$. Interpret your result.

EXERCISE 10

Suppose $X \sim \mathrm{N}(38.7,\ 8.2^2)$. Illustrate with a sketch and find k such that :

1. $\mathrm{P}(X \leqslant k) = 0.9$
2. $\mathrm{P}(X \geqslant k) = 0.8$

EXERCISE 11

Given that $X \sim \mathrm{N}(23,\ 5^2)$, find a such that :

1. $\mathrm{P}(X < a) = 0.378$
2. $\mathrm{P}(X \geqslant a) = 0.592$
3. $\mathrm{P}(23 - a < X < 23 + a) = 0.427$

EXERCISE 12

A university professor determines that no more than 80% of this year's History candidates should pass the final examination. The examination results were approximately normally distributed with mean 62 and standard deviation 13. Find the lowest score necessary to pass the examination.

EXERCISE 13

The length of a screw produced by a machine is normally distributed with mean 75 mm and standard deviation 0.1 mm. If a screw is too long it is automatically rejected. If 1% of screws are rejected, what is the length of the smallest screw to be rejected ?

EXERCISE 14

Pedro is studying Algebra and Geometry. He sits for the mid-year exams in each subject. Pedro's Algebra mark is 56%, and the class mean and standard deviation are 50.2% and 15.8% respectively. In Geometry he is told that the class mean and standard deviation are 58.7% and 18.7% respectively. What percentage does Pedro need to have scored in Geometry, to have an equivalent result to his Algebra mark ?

VI – Applications

Note. *We always need to convert to z-scores if we are trying to find an unknown mean μ or standard deviation σ.*

Example

An adult scallop population is known to be normally distributed with a standard deviation of 5.9 g. If 15% of scallops weigh less than 58.2 g, find the mean weight of the population.

Example

Find the mean and standard deviation of a normally distributed random variable X if $\mathrm{P}(X \leqslant 20) = 0.1$ and $\mathrm{P}(X \geqslant 29) = 0.15$.

EXERCISE 15

The weekly income of a bakery is normally distributed with a mean of \$6100. If 85% of the time the weekly income exceeds \$6000, what is the standard deviation ?

EXERCISE 16

The arrival times of buses at a depot are normally distributed with standard deviation 5 minutes. If 10% of the buses arrive before 3 :55 pm, find the mean arrival time of buses at the depot.

EXERCISE 17

The diameters of pistons manufactured by a company are normally distributed. Only those pistons whose diameters lie between 3.994 cm and 4.006 cm are acceptable.

1. Find the mean and the standard deviation of the distribution if 4% of the pistons are rejected as being too small, and 5% are rejected as being too large.
2. Determine the probability that the diameter of a randomly chosen piston lies between 3.997 cm and 4.003 cm.

EXERCISE 18

Circular metal tokens are used to operate a washing machine in a laundromat. The diameters of the tokens are normally distributed, and only tokens with diameters between 1.94 cm and 2.06 cm will operate the machine.

1. Find the mean and standard deviation of the distribution given that 2% of the tokens are too small, and 3% are too large.
2. Find the probability that at most one token out of a randomly selected sample of 20 will not operate the machine.

EXERCISE 19

Apples from a grower's crop were normally distributed with mean 173 grams and standard deviation 34 grams. Apples weighing less than 130 grams were too small to sell.

1. Find the proportion of apples from this crop which were too small to sell.
2. Find the probability that in a picker's basket of 100 apples, up to 10 apples were too small to sell.

Example

The time taken by students to complete a puzzle is normally distributed with mean 28.3 minutes and standard deviation 3.6 minutes. Calculate the probability that :

1. a randomly selected student took at least 30 minutes to complete the puzzle
2. out of 10 randomly selected students, at most half of them took at least 30 minutes to complete the puzzle.